ozone, ammonia, carbon monoxide, carbon dioxide, PM1.0, PM2.5 and PM10, and weather conditions such as ambient temperature, relative humidity, barometric pressure, wind direction, wind speed, noise and rainfall, are measured by the sensors in real time. The measurement data is digitised and logged, then sent wirelessly, through a cellular IP GPRS modem, to a cloud server via GPRS gateways. The connection can also be made using Wi-Fi or a LoRa network. The transmitted data is secured through AES 256-bit encryption. Data acquisition and analytics is performed on cloud using computational and predictive algorithms, and the results visualised on Aurassure's web interface in the form of charts, graphs, tables and other data representation methods. Aurassure also offers a mobile application that enables individual users to connect wirelessly to the cloud through their smartphone and check the data on-the-go. The system can work in the temperature range of -20°C to 55°C and relative humidity capacity between 5 per cent and 95 per cent.

Aurassure, when integrated with the web platform, provides a complete solution with features like remote device configuration and calibration. It can also be easily integrated with any third-party platform through standard API. The company has plans for further upgrades such as addition of more sensors and software features.

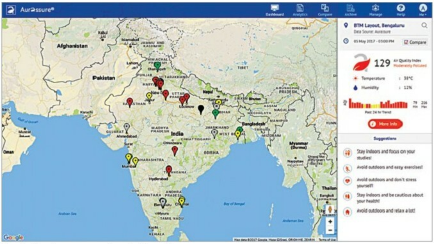

Fig. 3: The Aurassure online dashboard

The Aurassure team of Phoenix Robotix

Hitting the market with a bang

Launched earlier this year, Aurassure is currently in the initial marketing phase. Since the product is customised, it is priced on 'per device' basis. Usually, the pricing includes sensor requirements of the customer and subscription model for the online software platform. The product otherwise can cost up to one million rupees.

The company claims to have been receiving enquiries about Aurassure from different municipalities as well as companies across sectors. In fact, PSU major Steel Authority of India has implemented Aurassure in its Rourkela plant after it received several complaints from neighbouring townships and villages about emission of harmful pollutants. With permission from the regional office of State Pollution Control Board, Odisha, and Rourkela Steel Plant, Aurassure was installed near the fencing of the plant to enable fence-line monitoring.

Installed for three months till date, the system continuously measures pollutant emission from the plant. The collected data provides detailed insight into the operations that are majorly responsible for pollution and should be controlled. Acting on the data, the plant authorities were able to cut down chances of further emission. The solution also provided a pollution trend analysis, helping the plant authorities to plan their operations so as to minimise pollution to neighbouring residential places.

The way forward

Phoenix Robotix is primarily targeting municipalities of different cities, which invest heavily in the air quality monitoring infrastructure, along with different industries and environment consultants. The developers claim that, with Aurassure, industries will be able to perform fence-line monitoring of the environmental conditions of their work premises and around their locality.

"Aurassure has already undergone several successful POCs in different smart cities. It will significantly boost the number of deployments in various cities and provide a platform for accurate and useful micro-environmental data," claims Samantaray. EFY